Dharani Thallapally
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Professional Summary:

- Snowflake Developer with 5 years of expertise in designing, implementing, and optimizing cloud-based data warehousing solutions.
- Proficient in Snowflake's architecture, data sharing, and multi-cluster warehouse capabilities for high-performance analytics.
- Leveraged advanced features of Snowflake to design high-performance data warehousing solutions, enhancing scalability and operational efficiency for large datasets.
- Extensive experience in designing and optimizing logical and physical data models for Snowflake and Oracle databases.
- Demonstrated proficiency in AWS and Azure services, effectively integrating data pipelines and optimizing workflows across diverse cloud environments.
- Engineered robust real-time data ingestion pipelines using Snowpipe and AWS Kinesis, ensuring immediate data availability for analytics and reporting.
- Hands-on experience with tools like Informatica, Apache NiFi, AWS Glue, and Matillion for transforming and loading large datasets.
- Designed and implemented RESTful APIs with stringent security protocols (OAuth 2.0, JWT), facilitating secure, seamless data exchange between applications.
- Successfully synchronized data across AWS and Azure platforms, using tools like Azure Logic Apps and AWS DataSync to maintain data integrity.
- Proficient in writing complex SQL queries, stored procedures, and performance tuning in Snowflake.
- Implementing data masking, access controls, and managing secure data sharing across teams.
- Integrating Snowflake with data lakes Amazon S3, Azure Data Lake Storage for hybrid storage solutions.
- Working with Tableau, Power BI, and Looker to create dynamic dashboards and insights.
- Automating and orchestrating data pipelines using Python, PySpark, and Snowpipe.
- Implementing DevOps practices using Git, Jenkins, and Terraform for infrastructure as code (IaC).

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Technical Skills:

Category	Skills/Tools
Data Warehousing	Snowflake, Azure Data Factory, AWS RDS
Cloud Platforms	AWS (Kinesis, S3, Lambda, CloudFront), Azure
Data Ingestion	Snowpipe, AWS DataSync, RESTful APIs
Data Transformation	Python (Pandas, NumPy), SQL, Snowflake SQL functions
APIs & Security	RESTful API development, OAuth 2.0, JWT
ETL Processes	ETL pipelines (Snowflake, Azure Data Factory)
Analytics & BI	Tableau, SQL analytics, Snowflake analytical functions
Monitoring & CI/CD	AWS CloudWatch, Azure Monitor, Jenkins, GitHub Actions
Data Modeling	Dimensional modeling, OLAP cubes, logical and physical schema design
Performance Tuning:	Optimization for OLTP/OLAP databases, query tuning in Snowflake and Oracle.

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Professional Experience:

Client: USAA, San Antonio, Texas **Oct 2023 – Till Date**
Role: Snowflake Data Engineer

Description: As a Snowflake Developer at USAA, you will play a pivotal role in designing, building, and maintaining data-driven solutions that support the organization's mission. You will leverage your expertise in Snowflake's cloud-based data warehousing platform to develop scalable, efficient, and secure data solutions.

Responsibilities:

- Developed a real-time data pipeline leveraging Snowflake's capabilities to ingest data from various sources of transactional databases, APIs, and streaming services into a centralized data warehouse.
- Utilized AWS Kinesis for real-time data streaming and integrated it with Snowflake using Snowpipe, enabling the automatic loading of data as it arrives, thus supporting timely analytics and reporting.
- Utilized AWS S3 for scalable data storage, implementing data lakes to store structured and unstructured data. Developed automated scripts to regularly ingest data into Snowflake using Snowpipe for real-time loading.
- Configured AWS Lambda functions to trigger data ingestion processes based on events, such as new data uploads to S3, ensuring immediate availability for analysis.
- Created data flow pipelines to transform and cleanse data in Azure before loading it into Snowflake, ensuring high-quality datasets for analytical purposes.
- Developed and integrated RESTful APIs to facilitate real-time data exchange between Snowflake and external applications, enabling dynamic reporting and analytics for stakeholders.
- Established synchronization processes between AWS and Azure services to ensure data consistency across platforms, utilizing tools like Azure Logic Apps and AWS DataSync for seamless data transfer.
- Implemented endpoints for querying Snowflake data, allowing users to perform complex queries and retrieve analytical reports through user-friendly API calls, thereby enhancing data accessibility.
- Established robust authentication mechanisms, such as OAuth 2.0 and API keys, to ensure secure access to APIs and protect sensitive data during transmission.
- Implemented role-based access control (RBAC) within the API framework to restrict access based on user roles, ensuring compliance with data governance policies.
- Set up monitoring and logging for API performance using tools like AWS CloudWatch and Azure Monitor, allowing for real-time tracking of API usage, response times, and error rates.
- Developed complex SQL queries to extract, transform, and load (ETL) data within Snowflake, ensuring efficient data processing and retrieval for analytical reporting.
- Utilized advanced SQL techniques, such as window functions, common table expressions (CTEs), and subqueries, to enhance data analysis capabilities and optimize query performance.
- Leveraged Python to build data transformation scripts, automating ETL processes and enabling seamless data integration between various sources and Snowflake.
- Implemented data quality checks and validation logic using Python libraries (e.g., Pandas, NumPy) to ensure accuracy and reliability of data before loading into Snowflake.
- Developed and maintained RESTful APIs using Java Spring Boot, facilitating efficient communication between Snowflake and external applications, enhancing data accessibility for stakeholders.
- Implemented data serialization and deserialization using JSON and XML formats, ensuring smooth data interchange between systems.
- Created shell scripts and Python scripts to automate repetitive tasks, such as data extraction, loading, and monitoring, increasing operational efficiency and reducing manual effort.
- Utilized AWS Lambda with Python for serverless computing tasks, automating data ingestion and processing workflows in response to events, such as new data arrivals.
- Contributed to the development of CI/CD pipelines using tools like Jenkins or GitHub Actions to automate the deployment of scripts and APIs, ensuring rapid and reliable updates to production environments.
- Created fact and dimension tables to facilitate analytics on customer transactions, claims processing, and risk assessment, enhancing the organization’s ability to generate insights in real time.
- Designed and optimized physical and logical data models in Snowflake, leveraging dimensional modeling to support advanced analytics.
- Leveraged Snowflake’s multi-cluster architecture to ensure scalability and high availability of the data warehouse, accommodating fluctuating workloads without performance degradation.
- Developed robust ETL processes using Snowflake’s native SQL functions and stored procedures to transform raw data into meaningful insights, ensuring data quality and consistency across different datasets.
- Established data validation and cleansing mechanisms to identify and rectify data anomalies in real time, thereby maintaining high standards of data integrity for analytics purposes.
- Collaborated with cross-functional teams to align the data warehouse design with business objectives, ensuring that data models meet analytical needs and provide a 360-degree view of customer interactions.

Used Tools: AWS, Azure, API development, and SQL proficiency. Tools used encompass Snowflake, AWS services (S3, Kinesis, Lambda), Azure services, and CI/CD tools (Jenkins, GitHub Actions).

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Client: VanGuard, MI **Nov 2022 – Sep 2023**
Role: Snowflake Developer
Description: As a Snowflake Developer at Vanguard, I was responsible for designing, implementing, and maintaining data warehousing solutions using the Snowflake cloud-based data platform, with a specific focus on integrating Snowflake with Azure services.
Responsibilities:

- Designed and implemented scalable data warehousing solutions on the Snowflake platform, optimizing for performance and cost-effectiveness.
- Integrated Snowflake with various Azure services, including Azure Data Factory and Azure Data Lake, to streamline data ingestion and management processes.
- Developed and maintained ETL pipelines using Azure Data Factory to automate data workflows, ensuring efficient data transfer and transformation between source systems and Snowflake.
- Implemented Azure Data Governance best practices to ensure data integrity, compliance, and security, aligning with organizational policies and standards.
- Implemented efficient data loading strategies using Snowpipe for continuous data ingestion and bulk loading operations with COPY commands to handle large datasets seamlessly.
- A combination of Azure Synapse Analytics and Snowflake that provides a powerful and scalable data platform for building and deploying data-driven applications.
- Provided a unified analytics platform that integrates data warehousing, data lake, and machine learning capabilities
- Utilized built-in security features of Snowflake, such as role-based access control and data encryption, to safeguard sensitive data and maintain regulatory compliance.
- Leveraged Azure APIs and tools to facilitate seamless communication between Snowflake and external applications, enhancing data accessibility and usability for end-users.
- Provided programming access like Creating, managing, and querying databases, schemas, tables, and views.
- Allowed applications to connect to Snowflake using the ODBC standard, enabling data access from various programming languages and tools.
- Developed RESTful APIs using Azure Functions to enable real-time data access and manipulation, enhancing integration capabilities with external systems.
- Interacted with Azure SQL Database and with Snowflake for data replication and synchronization.
- Conducted regular performance tuning of Snowflake queries and data structures to enhance query performance and reduce processing times.
- Used business intelligence tool for creating interactive dashboards and reports.
- Conducted rigorous testing and debugging of ETL processes and SQL scripts, ensuring robustness and reliability in production environments.
- Collaborated with Business Analysts to define data requirements and implemented OLAP cubes for efficient multidimensional reporting.
- Collaborated with data engineers, analysts, and business stakeholders to gather requirements and design data solutions that meet business needs.
- Monitored data pipelines and warehouse performance using Snowflake's monitoring tools, proactively addressing issues to ensure high availability and reliability of data services.

Used Tools: Snowflake, Azure, Data Factory, Data Lake Storage, Purview, Snowpipe, COPY commands, Synapse, ODBC, SQL Database, Query, SSMS, BI Tools.

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Client: Cimpress, India **Jan 2020 – July 2022**
Role: Informatica Developer
Description: Cimpress's eCommerce domain enables businesses and consumers to easily create and order customized products through a user-friendly platform. It offers mass customization tools, integrated payment solutions, and online storefront capabilities. With a focus on enhancing the customer experience, Cimpress leverages data analytics to optimize marketing strategies and drive sales. This approach positions Cimpress as a leader in the eCommerce landscape for personalized products.
Responsibilities:

- I specialized in leveraging **AWS cloud services**, data analytics, and machine learning to build scalable, high-performance solutions that optimized the eCommerce platform, enabling real-time customization, seamless user experience, and data-driven decision making.

- Utilized **AWS Lambda** to create serverless functions that processed real-time data for image uploads, order processing without the need for maintaining dedicated servers.
- Set up **AWS S3** for secure and scalable storage of customer-uploaded images, design files, and other large assets required for product customization.
- Implemented **AWS RDS** for managing the platform's database with built-in backup, scaling, and failover support.
- Used AWS CloudFront as a Content Delivery Network **(CDN)** to distribute product images and assets globally, reducing latency and improving load times for end users.
- Configured **AWS Elastic Load Balancing (ELB)** to automatically distribute incoming traffic across multiple instances, ensuring platform reliability and uptime during high-demand periods.
- Monitored the platform's performance using **AWS CloudWatch**, setting up alerts for any anomalies and maintaining operational health by tracking resource utilization, request rates, and error counts.
- Integrated with front-end libraries like **React**, **Fabric.js**, or **Konva.js** to provide real-time customization capabilities.
- Used to expose **REST APIs** for customization-related requests.
- Analysed large datasets using **SQL and NoSQL databases** such as **AWS DynamoDB** and **MongoDB** for faster retrieval and indexing of data.
- Used **OAuth/OpenID Connect** for secure user authentication with third-party payment providers.
- Used **Tableau** for visualizing analytics and deriving insights from API data.
- Used **JWT (JSON Web Tokens)** for secure token-based authentication across services**.**
- Built predictive models using **AWS SageMaker** to forecast demand, optimize pricing strategies, and recommend personalized products.
- Processed **clickstream data** and user logs using tools like **AWS Kinesis** to capture real-time user interactions and inform business decisions.

Tools Used: AWS Lambda, AWS S3, AWS RDS, CloudFront, Elastic Load Balancing, CloudWatch, React, Fabric.js, Konva.js, AWS DynamoDB, MongoDB, OAuth/OpenID Connect, Tableau, JWT, SageMaker, Kinesis.

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Education:

Masters: Grand valley state university - Grand Rapids, Michigan, USA
Bachelors: CVR college of engineering – Hyderabad, India

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Certification:

Certified **Snowflake SNOW PRO**